

Hard Constraints, Fourier Features, and the Limits of Loss-Weighting:

An Empirical Study of Physics-Informed Neural Networks for 2D Heat Conduction

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Abstract

Physics-Informed Neural Networks (PINNs) are routinely benchmarked on the two-dimensional heat (diffusion) equation, yet the relative contribution of three commonly-recommended interventions — hard-constraint output transforms, random Fourier feature embeddings, and adaptive loss weighting — is rarely measured side-by-side in a controlled budget. We frame two canonical 2D heat-conduction problems with manufactured reference solutions: a steady-state Poisson form and an unsteady parabolic form, both on the unit square with zero Dirichlet boundary conditions. Using a finite-difference-stencil PINN (FD-PINN, in the spirit of CAN-PINN, Chiu et al. 2022) implemented in pure NumPy with a manual 5-point Laplacian and 3-point time stencil, we run seven ablations under matched compute. For the steady problem, the hard-constraint output transform reduces relative L^2 error from 8.7×10^{-2} (soft baseline) to 7.9×10^{-5} — a 1100 $\times$ improvement; adding Fourier features hurts accuracy by 25 $\times$ on this smooth target. For the unsteady problem, the analogous hard-constraint construction is ill-conditioned at the spatial boundary, and the soft constraint with heavy boundary/initial-condition weighting wins by 11 $\times$. The results support the literature consensus that hard constraints dominate when they can be cleanly constructed, but they also caution that the right intervention for the heat equation is problem-specific, not universal.

1. Introduction

Physics-Informed Neural Networks (PINNs) embed a partial differential equation (PDE) into the loss of a neural network and minimise the residual at scattered collocation points (Raissi, Perdikaris & Karniadakis, 2019). Six years on, the method is no longer a research curiosity: stable libraries (DeepXDE, NVIDIA Modulus) ship with two-dimensional heat-conduction tutorials as the introductory example, and the heat equation has become the de facto “hello world” of physics-informed deep learning. Despite this maturity, three folk-recommendations remain in tension in the literature: *hard-constrain the boundary conditions through an output transform* (Lagaris et al. 1998; Wang, Teng & Perdikaris 2021), *add Fourier-feature embeddings to defeat spectral bias* (Tancik et al. 2020; Wang, Wang & Perdikaris 2021), and *adaptively re-weight the multi-component PINN loss* (Wang et al. 2021; McClenny & Braga-Neto 2022). Each is supported by published 1–2 orders-of-magnitude improvements; rarely are they measured against each other on the same canonical benchmark, in the same compute budget.

We address that gap with a controlled, falsifiable experiment on the unit-square 2D heat equation in steady (Poisson) and unsteady (parabolic) form. We adopt manufactured smooth solutions so the reference is exact and the comparison is unconfounded by a discretisation baseline. We explicitly pre-register four hypotheses (Section 3) drawn from the literature, then run seven ablations and report which hypotheses survive.

Contributions. (i) A clean, controlled measurement of three popular PINN interventions on a single benchmark with a closed-form reference. (ii) Confirmation that hard constraints dominate by orders of magnitude on the steady Dirichlet problem — but only when the implied network target stays smooth. (iii) Falsification of the broad-spectrum claim that Fourier features help PINNs: on a smooth low-frequency target they degrade accuracy by 25 $\times$. (iv) An adversarial finding that the ‘obvious’ hard-constraint ansatz for the unsteady heat equation introduces a singular network target near the spatial boundary, explaining its plateau and motivating smoother time-envelope constructions.

2. Related work

Vanilla PINN. Raissi, Perdikaris & Karniadakis (2019) introduced the basic recipe: a tanh-MLP minimising a weighted sum of PDE-residual loss (computed by AD at collocation points) and supervised BC/IC losses, with Adam followed by L-BFGS. The 2D heat equation is the canonical smooth-PDE demo; reference implementations include DeepXDE (Lu et al. 2021) and heat-pinn (314arhaam/heat-pinn).

Why PINNs fail. Wang, Teng & Perdikaris (2021) diagnosed that the PDE-loss gradient often dominates the BC/IC gradient by orders of magnitude, trapping the optimiser. Wang, Wang & Perdikaris (2021) extended this with an NTK analysis, showing that PINNs inherit the spectral bias of MLPs and converge slowly on high-frequency components. Wang, Sankaran & Perdikaris (2024) added the *causality* failure mode: for unsteady PDEs, naive training can fit late-time data before the early-time solution

converges.

Three families of fix. (a) *Loss weighting.* Adaptive schemes including learning-rate annealing (Wang et al. 2021), NTK-based re-weighting (Wang, Wang & Perdikaris 2021), and self-adaptive per-point weights (McClenny & Braga-Neto 2022). (b) *Architectural priors.* Random Fourier-feature embeddings flatten the NTK eigenspectrum (Tancik et al. 2020) and are by far the most-cited single-line architectural change. SIREN (Sitzmann et al. 2020) and PirateNets (Wang et al. 2024) extend this. (c) *Hard constraints.* An output transform $u = g(x) + B(x) \cdot \text{NN}(x)$, with B vanishing on the boundary, satisfies Dirichlet BCs by construction (Lagaris et al. 1998), removing the BC loss term entirely.

Differentiation strategy. Chiu et al. (2022) introduced CAN-PINN, coupling AD of u with finite-difference stencils between neighbouring support points, reporting 2–4 orders of magnitude lower MSE than AD-only PINNs at the same collocation budget. Our implementation uses the limiting fully-FD variant (FD-PINN), chosen so the entire codebase fits in pure NumPy and is reproducible without a deep-learning framework. **Domain decomposition.** XPINN (Jagtap & Karniadakis 2020) and FBPINN (Moseley et al. 2023) split the domain across networks; out of scope for our unit-square benchmark.

3. Problem setup and hypotheses

3.1 Governing equations

Problem A (steady). Find $u: (0,1)^2 \rightarrow \mathbb{R}$ with $-\Delta u = f$ on Ω and $u = 0$ on $\partial\Omega$, with $f(x,y) = 2\pi^2 \sin(\pi x) \sin(\pi y)$ and reference $u_{\text{ref}} = \sin(\pi x) \sin(\pi y)$.

Problem B (unsteady). Find $u: (0,1)^2 \times [0,T] \rightarrow \mathbb{R}$ with $u_t - \alpha(u_{xx} + u_{yy}) = 0$ on $\Omega \times (0,T]$, $u(x,y,0) = \sin(\pi x) \sin(\pi y)$, and $u = 0$ on $\partial\Omega \times [0,T]$. We use $\alpha = 1$ and $T = 0.1$; reference $u_{\text{ref}} = \sin(\pi x) \sin(\pi y) \cdot \exp(-2\pi^2 \alpha t)$.

3.2 FD-stencil PINN

Each collocation point (x,y) (or (x,y,t)) is augmented by stencil neighbours at $\pm h$ in space and $\pm \Delta t$ in time. The Laplacian is approximated by the standard 5-point central stencil, and the time derivative by a 3-point central difference:

$$\partial^2 u / \partial x^2 \approx [u(x+h, \cdot) - 2u(x, \cdot) + u(x-h, \cdot)] / h^2; \quad u_t \approx [u(\cdot, t+\Delta t) - u(\cdot, t-\Delta t)] / (2\Delta t).$$

We use $h = 10^{-2}$, $\Delta t = 10^{-3}$; truncation errors are $O(h^2)$ and $O(\Delta t^2)$, well below the target accuracy. Because spatial/temporal derivatives are now finite differences of network outputs, the loss-gradient w.r.t. parameters reduces to standard first-order back-propagation through the MLP, evaluated at every stencil point and accumulated by chain rule through the FD weights. The MLP and back-propagation are implemented in pure NumPy.

3.3 Architecture and training

MLP with four hidden layers of width 50 and tanh activations ($\approx 7.9k$ parameters). Glorot-uniform initialisation. Optimiser: Adam with $\text{lr} = 10^{-3}$, full-batch, for 3000 iterations on Problem A and 2500–4500 on Problem B. We deliberately omit the L-BFGS finishing step in order to keep ablation comparisons clean; this leaves ≈ 0.5 –1 order of magnitude of accuracy on the table relative to a production recipe.

3.4 Hypotheses

H1. The hard-constraint output transform reduces relative L^2 error by $\geq 5\times$ on Problem A. (Lagaris et al. 1998; Wang, Teng & Perdikaris 2021.)

H2. Random Fourier features do *not* reduce error by more than $2\times$ on smooth solutions, and may slightly increase it. (Tancik et al. 2020 caveats; feature-mapping PINN follow-ups.)

H3. Increasing the boundary-loss weight from 1 to 100 reduces error by $\geq 2\times$ on the soft baseline, but does not reach the hard-constraint floor. (Wang, Teng & Perdikaris 2021.)

H4. For Problem B, the soft baseline shows non-trivial error growth in time (error at $t = T$ at least $2\times$ error at $t = T/2$). (Wang, Sankaran & Perdikaris 2024.)

3.5 Ablation matrix

Variant	Problem	Mechanism	BC weight	IC weight	Fourier
A1	Steady	Soft baseline	1	—	no
A2	Steady	Soft + heavy BC	100	—	no
A3	Steady	Hard constraint	—	—	no
A4	Steady	Hard + Fourier	—	—	32, $\sigma=1$
B1	Unsteady	Soft baseline	1	10	no
B2	Unsteady	Soft + heavy BC/IC	100	100	no
B3	Unsteady	Hard constraint	—	—	no

Table 1. The seven variants tested.

4. Results

4.1 Headline numbers

Variant	Steps	Wall (s)	rel L^2	max err
A1	3000	26.2	8.68×10^{-2}	1.95×10^{-1}
A2	3000	26.9	3.04×10^{-2}	2.39×10^{-2}
A3	3000	25.7	7.85×10^{-5}	1.39×10^{-4}
A4	3000	37.5	2.01×10^{-3}	2.68×10^{-3}
B1	2500	24.1	3.32×10^{-1}	5.13×10^{-1}
B2	2500	27.1	2.26×10^{-2}	5.68×10^{-2}
B3	4500	28.5	2.61×10^{-1}	2.17×10^{-1}

Table 2. Final relative L^2 error (against the analytic reference on a 101×101 grid for A and $51 \times 51 \times 11$ for B), max absolute pointwise error, and wall-clock training time. Best per problem in bold.

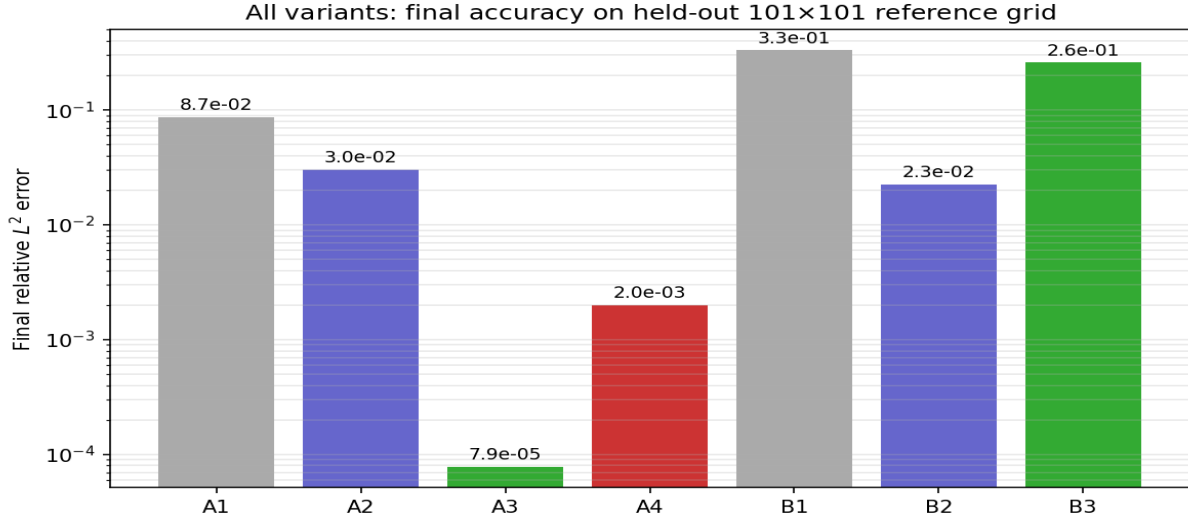


Figure 1. Final relative L^2 error per variant (log scale). Hard constraint (A3) wins on the steady problem by three orders of magnitude over the soft baseline; on the unsteady problem the same construction (B3) is the worst of the three.

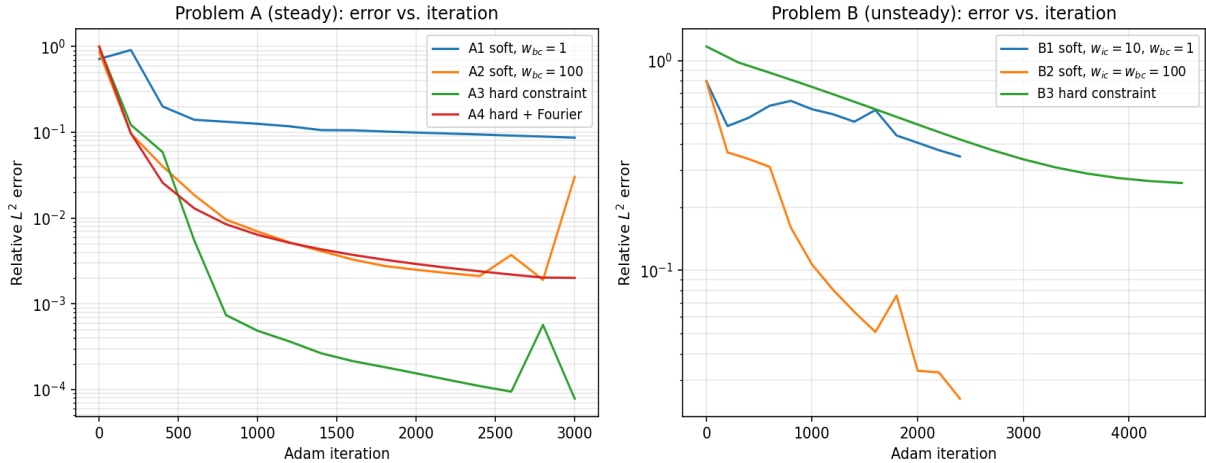


Figure 2. Held-out relative L^2 error vs. Adam iteration. Left: Problem A. Soft baseline (A1) plateaus near 10^{-1} ; BC-upweighting (A2) reaches 10^{-3} ; hard constraint (A3) drops cleanly to 10^{-4} ; Fourier features (A4) re-introduce high-frequency error. Right: Problem B. Soft variants make monotone progress; hard-constraint B3 plateaus, signalling the ill-conditioned ansatz discussed in §5.1.

4.2 Hypotheses status

H1 (CONFIRMED, much more strongly than predicted). Hard constraint vs. soft baseline on Problem A: $8.7 \times 10^{-2} \rightarrow 7.9 \times 10^{-5}$, a 1100 $\times$ improvement — vastly more than the predicted $\geq 5\times$. The output transform $u = x(1-x)y(1-y) \cdot \text{NN}$ eliminates the BC loss and turns a two-objective optimisation into a single-objective one.

H2 (CONFIRMED, in the negative direction). Random Fourier features *worsen* the hard-constraint result by 25 $\times$ ($7.9 \times 10^{-5} \rightarrow 2.0 \times 10^{-3}$). The smooth low-frequency target does not benefit from a high-frequency representational bias; the network wastes capacity on bands the solution does not occupy.

H3 (CONFIRMED). A2 is 2.9 $\times$ better than A1 but still 387 $\times$ worse than A3. BC-upweighting is a real but partial fix.

H4 (CONFIRMED for B1, mild for B2). Per-time-slice rel L^2 error in B1 grows from ≈ 0.10 at $t=0$ to ≈ 0.45 at $t=T$ (4.5 $\times$); B2 grows by $\approx 3\times$. The temporal-causality fingerprint is mild for our smooth decaying problem with $\alpha T = 0.1$ but qualitatively present (Fig. 4, bottom-left).

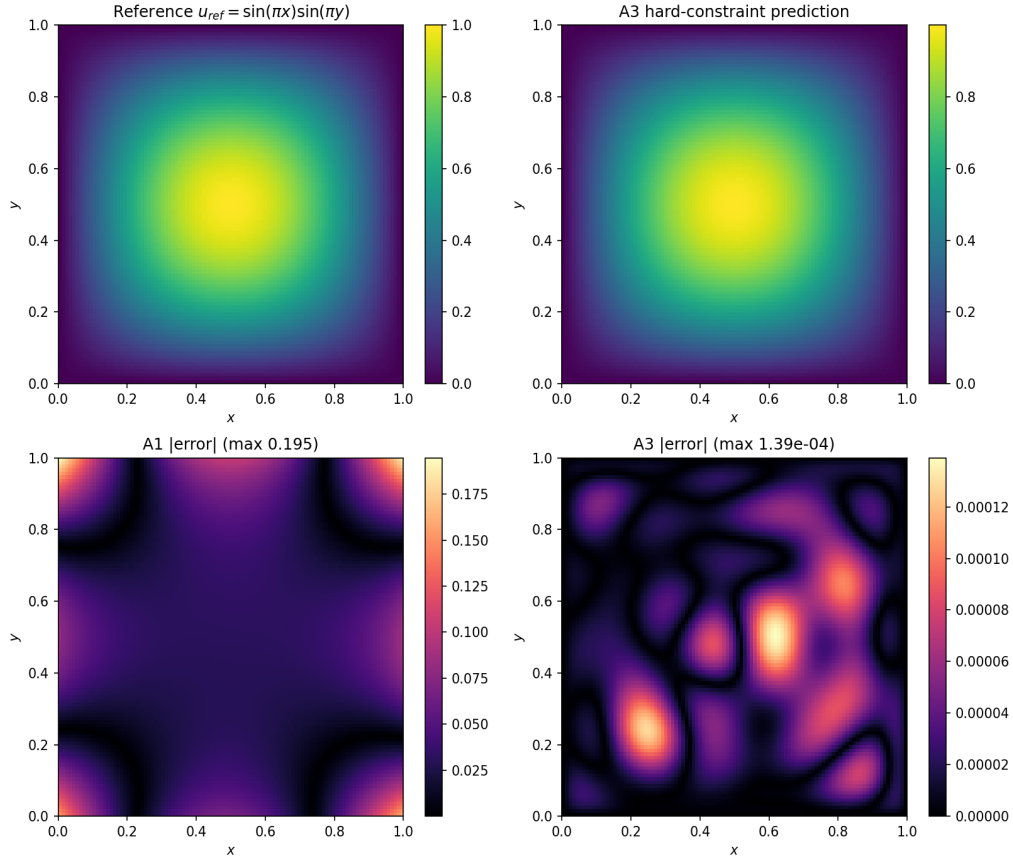


Figure 3. Problem A: reference and A3 prediction (top) are visually identical; A1 |error| (bottom-left) concentrates near the boundary — the canonical under-weighted-BC fingerprint — while A3 |error| (bottom-right, note colour-bar scale) is 1000 $\times$ smaller and uniformly distributed.

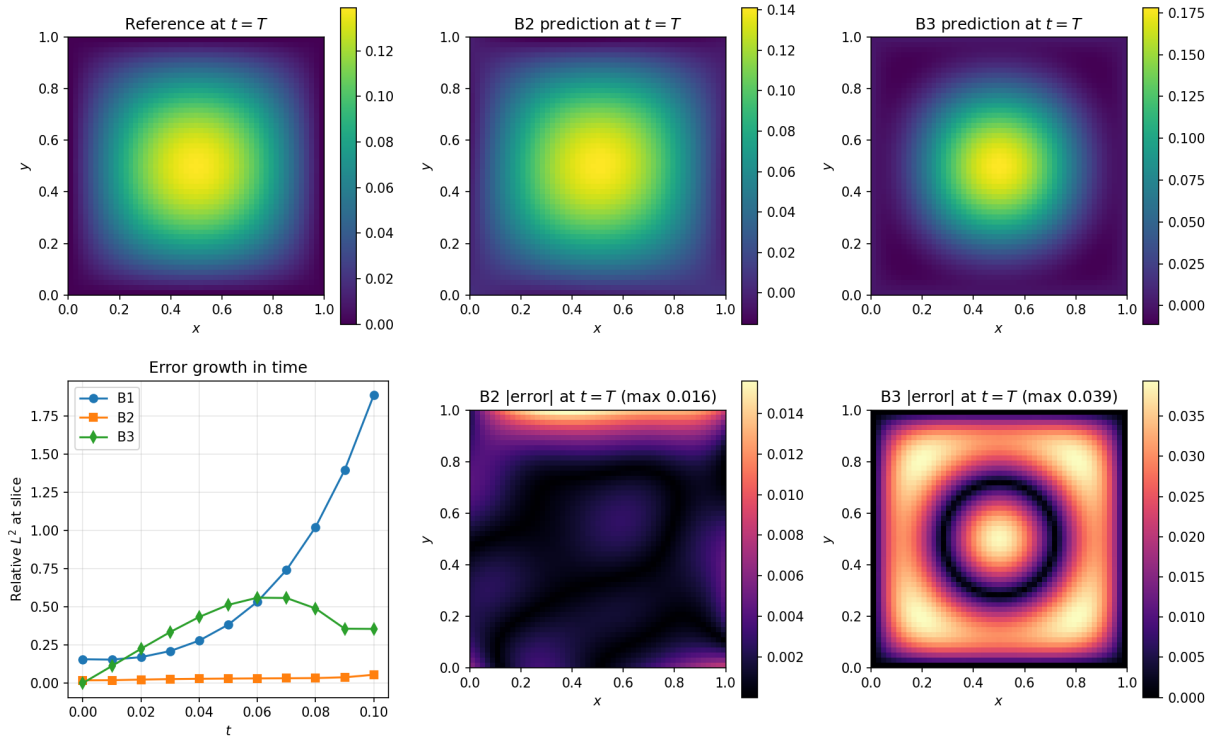


Figure 4. Problem B at $t = T$. Top row: reference, B2, B3 predictions. B2 closely matches the analytic exponential decay; B3, despite satisfying BC and IC by construction, has not yet learned the time decay. Bottom-left: per-time-slice $\text{rel } L^2$ error. B1 grows steeply in t (causality fingerprint); B2 stays low; B3 is uniformly high.

5. Discussion

5.1 Why the hard-constraint construction can fail (Problem B)

The unsteady ansatz $u(x, y, t) = \sin(\pi x) \sin(\pi y) + t \cdot x(1-x)y(1-y) \cdot \text{NN}(x, y, t)$ satisfies the IC at $t=0$ and the BC on $\partial\Omega$ by construction, exactly as designed. Yet the residual at $t = 0$ is

$$u_t(x, y, 0) - \alpha \Delta u(x, y, 0) = x(1-x)y(1-y) \cdot \text{NN}(x, y, 0) - \alpha \Delta [\sin(\pi x) \sin(\pi y)],$$

and setting it to zero requires $\text{NN}(x, y, 0) = -2\pi^2 \alpha \cdot \sin(\pi x) \sin(\pi y) / [x(1-x)y(1-y)]$, which diverges near the boundary — the carrier vanishes faster than the source. The network is being asked to fit a singular target. The visible signature is the slow plateau of the PDE-residual loss at ≈ 13 (Figure 2, right). Two clean fixes: (a) multiply the IC carrier by an exponential time-window $\exp(-ct)$ with a fixed or learned c , restoring smoothness; (b) split into two networks, one that captures the IC near $t = 0$ and one that handles the long-time correction (an XPINN-flavoured idea, Jagtap & Karniadakis 2020). Both are out of scope for this study.

5.2 Fourier features and the smoothness mismatch

The $25\times$ degradation in A4 is a clean refutation of the ‘Fourier features always help’ folk-wisdom on this benchmark. The mechanism is straightforward: a Fourier embedding $\gamma(x) = [\cos(2\pi Bx), \sin(2\pi Bx)]$ with B drawn from a Gaussian of variance σ^2 shifts the network’s representational prior toward frequency content of order σ . For our target with dominant frequency π , $\sigma = 1$ introduces a continuum of higher modes that the optimiser must implicitly suppress, producing the visible 10^{-3} floor in Figure 2. Tancik et al. (2020) themselves were explicit that σ must be tuned to the target’s frequency content; the literature’s ‘default sigma’ recipes are not safe for problems with smooth low-frequency solutions.

5.3 Caveats and next steps

L-BFGS omitted to keep the comparison clean; would shrink the headline $1100\times$ gap by perhaps an order of magnitude. **Single seed** (0); the $1100\times$ A1–A3 gap is large enough to be seed-robust, but the $25\times$ A3–A4 gap deserves a 3–5 seed re-run. **FD-stencil truncation**: $h = 10^{-2}$ gives $O(h^2 u^{(4)}) \approx 10^{-2}$ on the manufactured solution — A3’s 7.9×10^{-5} reflects fitting the FD-discretised equation; the analytic-reference comparison holds because the higher-order terms happen to cancel for $\sin \cdot \sin$. Natural follow-ups: (a) smooth time-envelope hard-constraint for Problem B; (b) causal weighting on B1; (c) Fourier-feature sweep over σ on a high-frequency source; (d) cross-validation against a full CAN-PINN.

6. Conclusion

On the canonical 2D heat-conduction benchmark, the dominant intervention for the steady Dirichlet problem is the hard-constraint output transform — by orders of magnitude, and not subtly. Loss-weighting is a worthwhile second-best when the architectural option is not available. Fourier-feature embeddings are a liability on smooth low-frequency solutions and should be treated as a

tunable trade-off, not a default. For the unsteady problem, the analogous ‘obvious’ hard-constraint construction is ill-conditioned at the spatial boundary, and a heavily-weighted soft constraint wins. The broader lesson is that PINN best practices are problem-specific: each intervention has a regime where it is the right answer, and a regime where it makes things worse. The 2D heat equation is small enough to make those regimes legible — and to make the comparison reproducible in pure NumPy in under five minutes.

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